UNIVERSITY OF LEICESTER

CO2124

Muhammad Iftikhar

Lecture 7:
Mobile App Development

Today's lecture includes,

- Android Architecture
- Android Components
- Activities Lifecycle
- Android Development
- More in CO3102 Mobile and Web Applications
  - In-depth study of interchange formats
  - Native vs Hybrid apps

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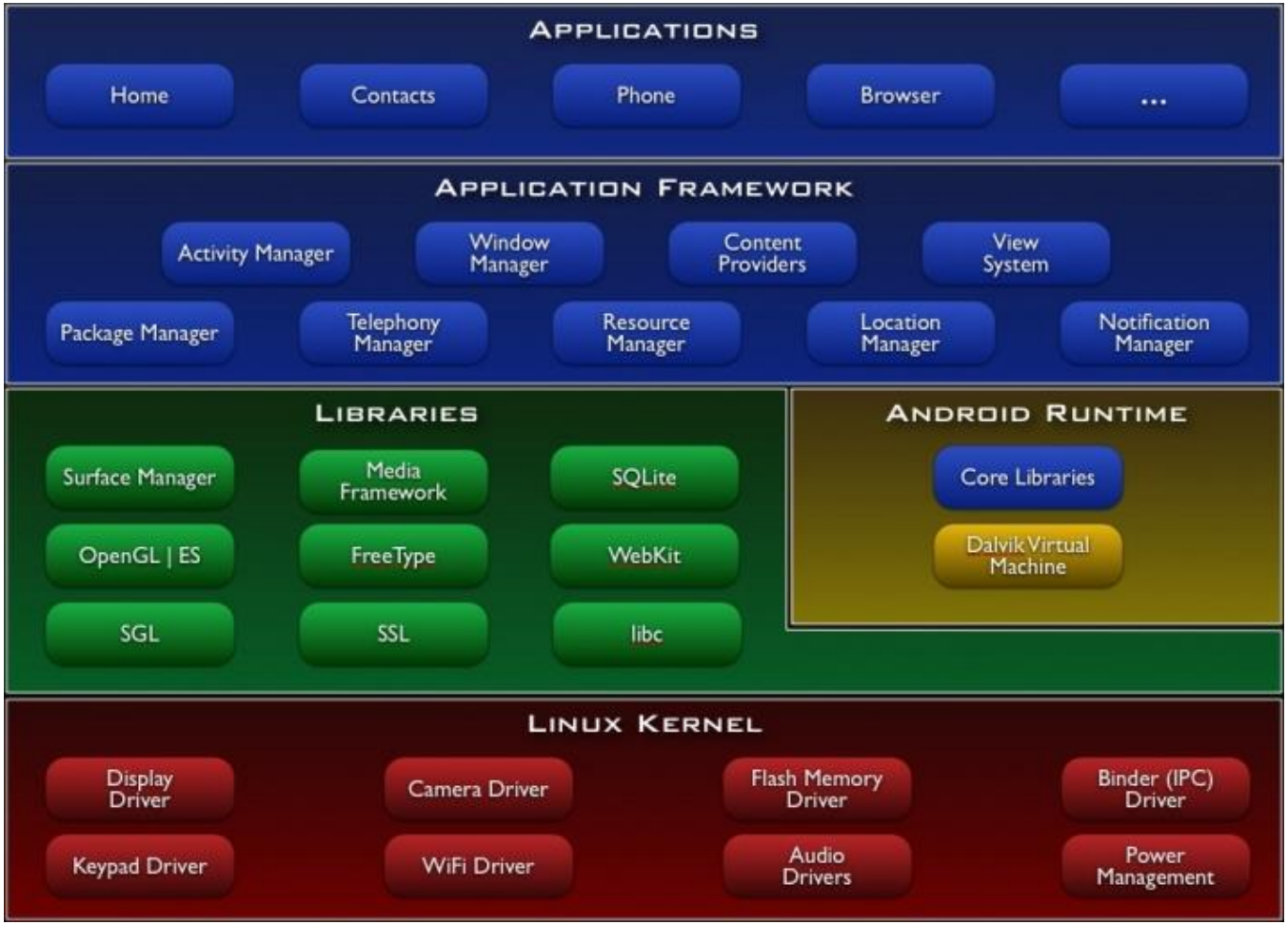
Wide Range of Devices

Android Architecture



https://elinux.org/Android_Architecture

Linux Kernel & Android Runtime

- **Linux Kernel:** Core OS services; Security, multi-threading, memory management, process management, etc.
  - Abstraction layer between software and hardware

- **Android Runtime:** Core Java libraries
- Every Android app runs in its own process, with its **own instance** of the **Dalvik virtual machine**
  - The Dalvik VM executes files in Dalvik Executable (.dex) format, optimised for minimal memory footprint
  - Dalvik runs classes compiled by a Java compiler which have been transformed into the .dex format

Libraries

C/C++ libraries: Exposed through the Android Application Framework

System C library: Standard C system library (libc) tuned for embedded Linux-based devices

Media Libraries: Support recording and playback of popular image, audio and video formats, e.g., JPG, PNG, MPEG4, H.264, MP3, AAC

LibWebCore: Web browser engine which powers both the Android browser and an embeddable web view

3D libraries: For hardware 3D acceleration

SQLite: A powerful, lightweight relational database engine

(and more)

Application Framework

- Provides access to device hardware, location information, background services, alarms, notifications, etc
- Designed for component reuse and interplay between apps; apps can publish and reuse capabilities (subject to security constraints)
- Underlying all apps is a set of services and systems, including:
  - **Views,** used to build apps, e.g., lists, grids, text boxes, buttons, browser
  - **Content Providers** enable apps to interchange data with other applications (e.g. Contacts)
  - **Resource Manager** providing access to non-code resources such as localised strings, images, and layout files
  - **Notification Manager** enables apps to display custom alerts
  - **Activity Manager** manages app lifecycle and navigation

Applications

- Written (mostly) in Java
- Compiled Java code, data and resource files are bundled into an Android **.apk** package
- **.apk** files are used to **distribute** and **install** apps on mobile devices
- Each app runs in its own Linux process. Android starts the process when any of the app's code needs to be executed, and shuts down the process when it's no longer needed and system resources are required by other apps
- Each process runs on its own Dalvik VM, so application code runs independently from the code of all other apps
- **Device Adaptation:** different resources for different devices, e.g., different layouts for different screen sizes, feature enabled / disabled based on hardware specs (such as fingerprint reader)

Applications

- Apps can **share components**, e.g., an app requires to display a scrolling list of images and another app provides a suitable scroller; this can be arranged on-demand
- Apps don't have a single entry point, i.e., **no main() function**
  - Apps are built as a combination of components
  - The system starts an app’s process and instantiate the necessary Java objects when any component of the app is requested
  - Other components, such as broadcast receivers and services, allow your app to perform background tasks without a UI

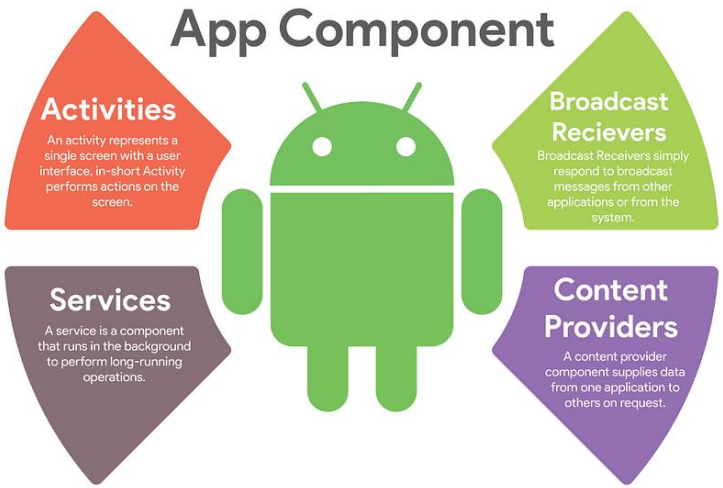
Main Components of an Android App

Activities

Services

Broadcast Receivers

Content Providers



Activities

- Primary element for user interaction, e.g., a single screen where the app draws its UI
- Implements a focussed task that the user can do
- The **main activity** is executed when the user taps your app's icon
- An activity is declared in the **AndroidManifest** XML file and implemented as a subclass of the **Activity** class

```
<activity
    android:name=".MainActivity"
    android:label="CO2103-FirstApp"
    android:theme="@style/AppTheme.NoActionBar">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNC
    </intent-filter>
</activity>
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
```

Activities Lifecycle

- When an Android app is first started, the main activity is created. The activity then goes through 3 states before it is ready to serve the user: Created, started and resumed.
- If an activity A opens another activity B, then activity A will be paused. This means that activity A goes into the paused state. When the user clicks the back button and returns to activity A, activity A returns to the resumed state.
- If the user returns to the home screen of the Android device, all activities will be paused and then stopped. If the user then returns to the app, the activities will go through the started and then resumed states. If the Android device needs the memory that the Android app occupies in the device's memory, then it may completely destroy the app, meaning the Android activities go into the destroyed state.

Activities Lifecycle

onCreate(): must be implemented; fires when system creates activity

- In the case of WhatsApp, when you open the app, onCreate() would be called.

onStart(): executes after **onCreate**, right before the activity becomes visible to the user to interact with it (**Started** state)

- onStart() might involve tasks like fetching recent chat messages or loading contact lists.

onResume(): invoked just before the activity starts interacting with the user (again). The **onPause()** callback always follows **onResume()**

- this might involve tasks like establishing network connections for real-time messaging or starting animations for user interface elements.

onPause(): Invoked when the activity loses focus and enters the **Paused** state; once it finishes executing, the next callback is either **onStop()** or **onResume()**

- onPause() might be called when the user receives a notification from another app while they are still in conversation. Whats app might handle this event by pausing any ongoing tasks, such as playing voice messages, but still keeping the activity alive in the background.

Activities Lifecycle

onStop(): the activity is no longer visible to the user; followed by **onRestart()** or **onDestroy()**

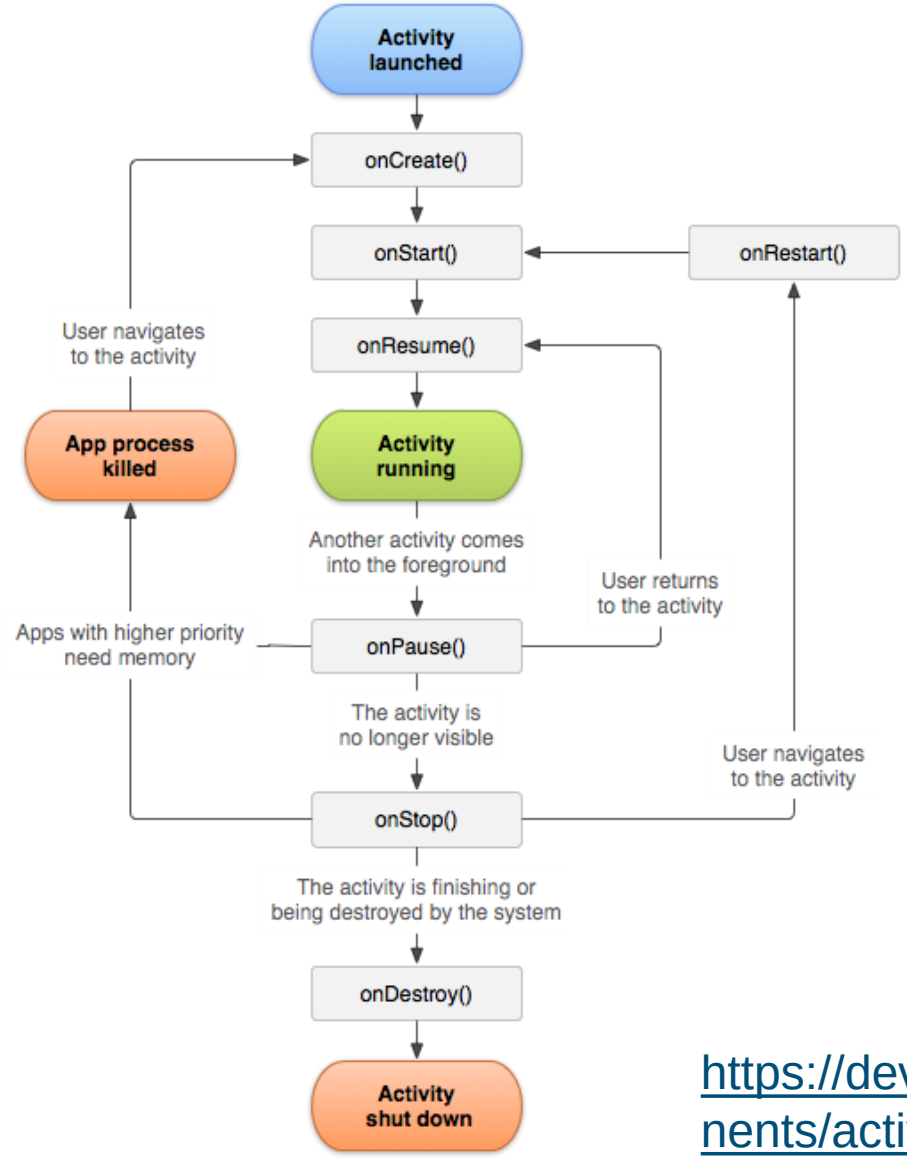
- When WhatsApp is no longer visible to the user, onStop() is triggered. This could happen if the user switches to another app entirely or goes back to the home screen.

onRestart(): when an activity in the **Stopped** state is about to restart; always followed by **onStart()**

- If the user returns to WhatsApp after it has been stopped but not destroyed, onRestart() is called.

onDestroy(): invoked before an activity is destroyed

- WhatsApp is being closed or if the system needs to free up memory, onDestroy() is called.



Note that the states always follows the arrows in the diagram. An activity cannot jump from created to resumed directly. An activity will always go through created, started and resumed in that sequence. An activity may change from resumed to paused and back, and from paused to stopped and back to started, but never from stopped directly to resumed.

https://developer.android.com/guide/components/activities/activity-lifecycle

Services, Broadcast Receivers, and Content Providers

Services

- Run in the background
- Support long-running operations, e.g., downloading media
- Support interaction with remote processes, e.g., email client
- A service is implemented as a subclass of **Service**

Broadcast Receivers

- Listen for and respond to events
- Events broadcast using the intent class; e.g., messaging App

Content Providers

- Multiple apps can store/share data
- Database-style interface

See you in the next lecture…